

Question			Marking details	Marks available				Maths	Prac																		
				AO1	AO2	AO3	Total																				
2	(a)	(i)	Magnitude of vertical force = 2.0×10^{-4} (N) (1) [Direction not required here] Application of Pythagoras and correct overall magnitude determined i.e $R^2 = (2.0 \times 10^{-4})^2 + (5.0 \times 10^{-4})^2$ $R = 5.4 \times 10^{-4}$ (N)[accept 5.38 or 5.39] (1) No ecf $\theta = 21.8^\circ$ [tolerance of rounding errors] below horizontal [allow 112° , 22° South of E...] stated or clearly shown in diagram [accept 68.2° to the vertical stated or shown] ecf on R (1)		3		3	3																			
		(ii)	Air resistance and force due to gravity [or weight] are equal [or air resistance = 6.0×10^{-4} N] hence no resultant force [accept forces balanced /cancel / no acceleration]	1			1																				
	(b)	(i)	Subtract ...0.05 [from readings (of time)] /the time delay / ...it			1	1		1																		
		(ii)	<table border="1"><tr><td>Drop height h, (m)</td><td>0.40</td><td>0.80</td><td>1.20</td><td>1.60</td><td>2.00</td></tr><tr><td>Corrected time t, (s)</td><td>0.27</td><td>0.41</td><td>0.48</td><td>0.58</td><td>0.64</td></tr><tr><td>Corrected time squared t^2, (s²)</td><td>0.07(3)</td><td>0.17</td><td>0.23</td><td>0.34</td><td>0.41</td></tr></table> All values of t^2 calculated correctly (1) [award this mark, even if sig figs incorrect] To 2 sig. fig. [Allow 1 sf on first answer] [see table] (1)	Drop height h , (m)	0.40	0.80	1.20	1.60	2.00	Corrected time t , (s)	0.27	0.41	0.48	0.58	0.64	Corrected time squared t^2 , (s ²)	0.07(3)	0.17	0.23	0.34	0.41		2		2	2	2
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		(iii)	$x = ut + \frac{1}{2}at^2$ identified (1) Explanation that: [$x = h$], $u = 0$ and $a = g$ [or by implication] (1) No algebra required	1	1		2																				

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		(iv)	Suitable scale and both axes labelled correctly with appropriate units: [drop] height [h]/ m and time squared [t^2]/ s^2 [or (t / s)²] (1) Allow ecf from table All 5 points plotted correctly $\pm \frac{1}{2}$ small square division (2) If 4 points plotted correctly $\pm \frac{1}{2}$ small square division (1) If 3 or less points plotted correctly $\pm \frac{1}{2}$ small square division (0) Appropriate line of best fit [through origin] (1) ecf gradient $= \frac{1.92}{0.4}$ $= 4.8$		4		4	4	4
		(v)	Suitable triangle shown on graph [or two points on line implied in calculation] with $\Delta h \geq 1.0$ m [or two appropriate points shown on line] See above (1) Gradient calculated correctly [Accept 4.6 to 5.0] (1) $g = 2 \times \text{gradient}$ [Allow ecf] (1) 2nd and 3rd mark can be awarded even if first mark withheld. [Allow final mark for correct answers using data points rather than gradient]			3	3	2	3

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	(c)		Straight line / $h \propto t^2$ / linear graph (1) Through [or close to] origin (1) g close to accepted value / 9.81 or low degree of scatter / points close to line of best fit [accept relevant comment based upon candidate's graph] (1)			3	3		3
			Question 2 total	2	10	7	19	11	13